

E-commerce Cohort Retention Analysis

Portfolio presentation | Product analytics case study

CONTEXT

Product team suspects that many newly observed users may not be returning consistently

BUSINESS QUESTION

How can we validate newly observed user retention and improve retention outcomes?

EXECUTIVE SUMMARY

Dataset	RetailRocket public dataset with 2.7M+ user-level events
Key finding	Largest retention decline occurs between D1 and D7, and remains broadly consistent across mature cohorts
Action	Use early browsing breadth as an engagement signal to encourage early exploration and re-engage low-breadth users

How retention was measured



Cohort anchor

First observed event date

first_seen



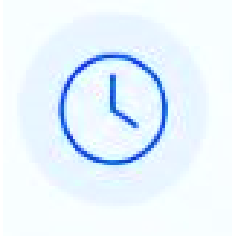
Retention metric

Exact-day any-event return

view

addtocart

transaction



Checkpoints

D1, D7, D14, and D30
measured using full 24-hour
day buckets



Segmentation

First-3-day browsing breadth
1 item / 2 or 3 items / ≥ 4 items

Note: first_seen is a proxy for signup, and retention is measured only for users with enough observation time.

The largest retention drop happens between D1 and D7

Overall exact-day any-event retention by checkpoint

Checkpoint	Mature users	Retained users	Exact-day retention rate	Drop from previous checkpoint
D1	1,366,774	21,813	1.596%	-
D7	1,328,821	6,756	0.508%	 -1.088 pp
D14	1,262,817	3,730	0.295%	 -0.213 pp
D30	1,115,048	1,651	0.148%	 -0.147 pp

* *pp* = percentage points

* *Mature users* means users whose first-seen date is early enough that the dataset has enough observation time to measure that checkpoint.

Exact-day retention generally declines from D1 to D30, especially drops sharply after Day 1 and before Day 7. This suggests **an early lifecycle window to investigate**.



Next: *Is the D1–D7 drop mainly driven by a few unusual cohorts?*

The D1–D7 drop is broadly consistent across cohorts

Weekly exact-day retention by first-seen cohort

Cohort Week	Cohort Users	D1	D7	D14	D30
27 April 2015	7,865	3.65%	1.46%	1.17%	0.61%
04 May 2015	71,389	2.08%	1.04%	0.61%	0.32%
11 May 2015	76,662	1.97%	0.67%	0.37%	0.17%
18 May 2015	78,427	1.69%	0.50%	0.33%	0.15%
25 May 2015	73,231	1.60%	0.54%	0.27%	0.16%
01 June 2015	72,714	1.69%	0.54%	0.32%	0.15%
08 June 2015	66,586	1.57%	0.53%	0.28%	0.18%
15 June 2015	69,757	1.66%	0.57%	0.34%	0.14%
22 June 2015	68,400	1.67%	0.50%	0.31%	0.14%
29 June 2015	68,147	1.55%	0.55%	0.26%	0.13%
06 July 2015	83,944	1.45%	0.43%	0.25%	0.11%
13 July 2015	80,563	1.34%	0.46%	0.24%	0.10%
20 July 2015	88,261	1.60%	0.38%	0.23%	0.11%
27 July 2015	73,190	1.44%	0.44%	0.21%	0.14%
03 August 2015	67,081	1.46%	0.40%	0.25%	0.12%
10 August 2015	64,291	1.41%	0.38%	0.22%	0.13%
17 August 2015	65,264	1.54%	0.40%	0.29%	0.09%
24 August 2015	65,333	1.49%	0.36%	0.28%	
31 August 2015	62,329	1.53%	0.50%	0.22%	
07 September 2015	71,205	1.55%	0.38%		
14 September 2015	32,941	1.05%			

Most mature weekly cohorts follow a similar pattern: D1 is highest, followed by lower D7, D14, and D30 retention. This makes the **early drop-off pattern** found in the overall table **more reliable**.

Also strengthens the case that the **D1–D7 window is worth investigating** for early retention actions

Note: first and last cohort weeks are boundary cohorts and excluded from interpretation.

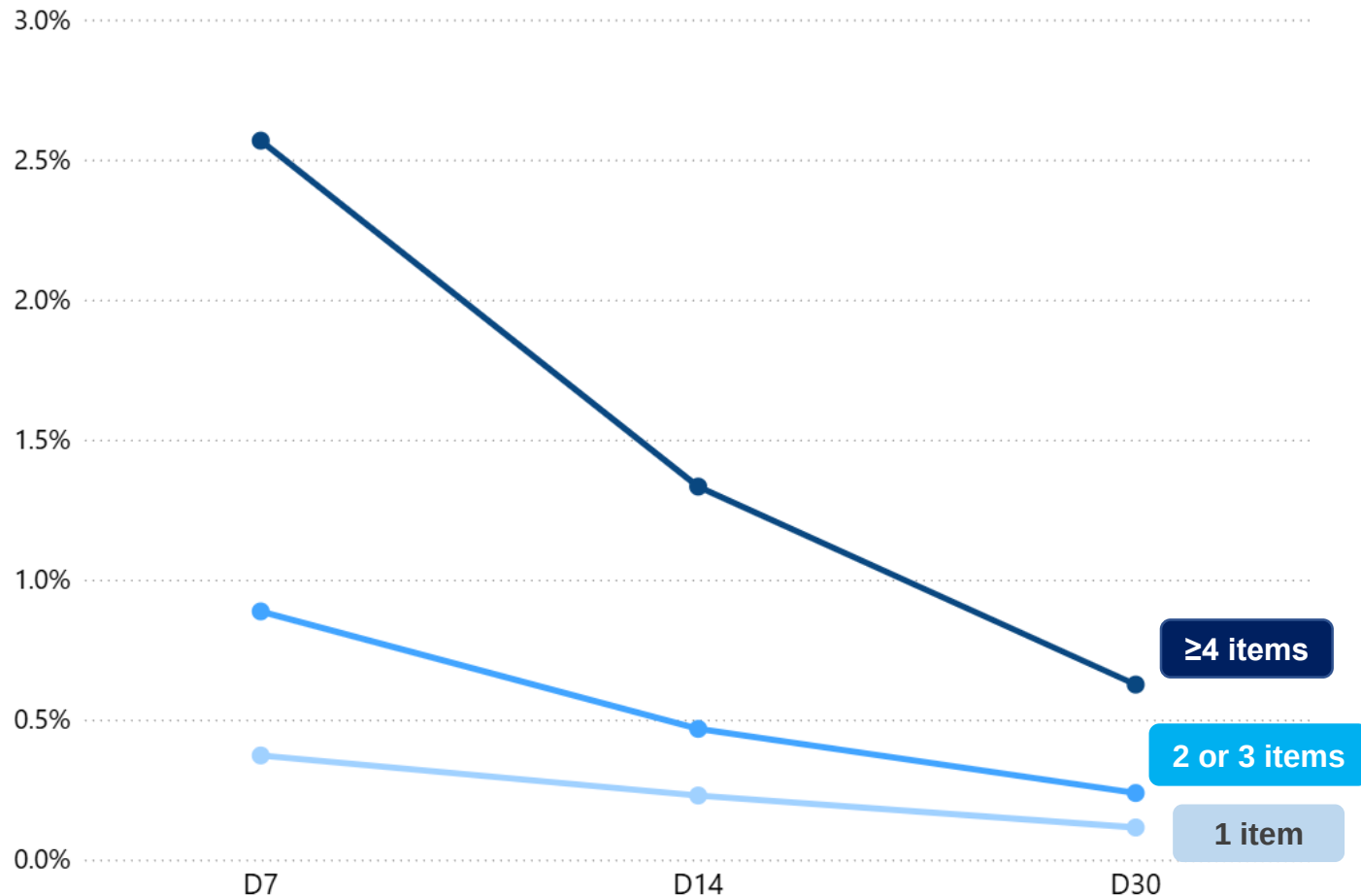


Next: Can early user behavior help distinguish users with stronger vs weaker retention?

Broader early browsing is associated with stronger retention at D7, D14, and D30

i Segments are based on the number of distinct items that users interacted with during the first 3 days: 1 item, 2 or 3 items, and equal or more than 4 items.

Exact-day retention by early browsing breadth

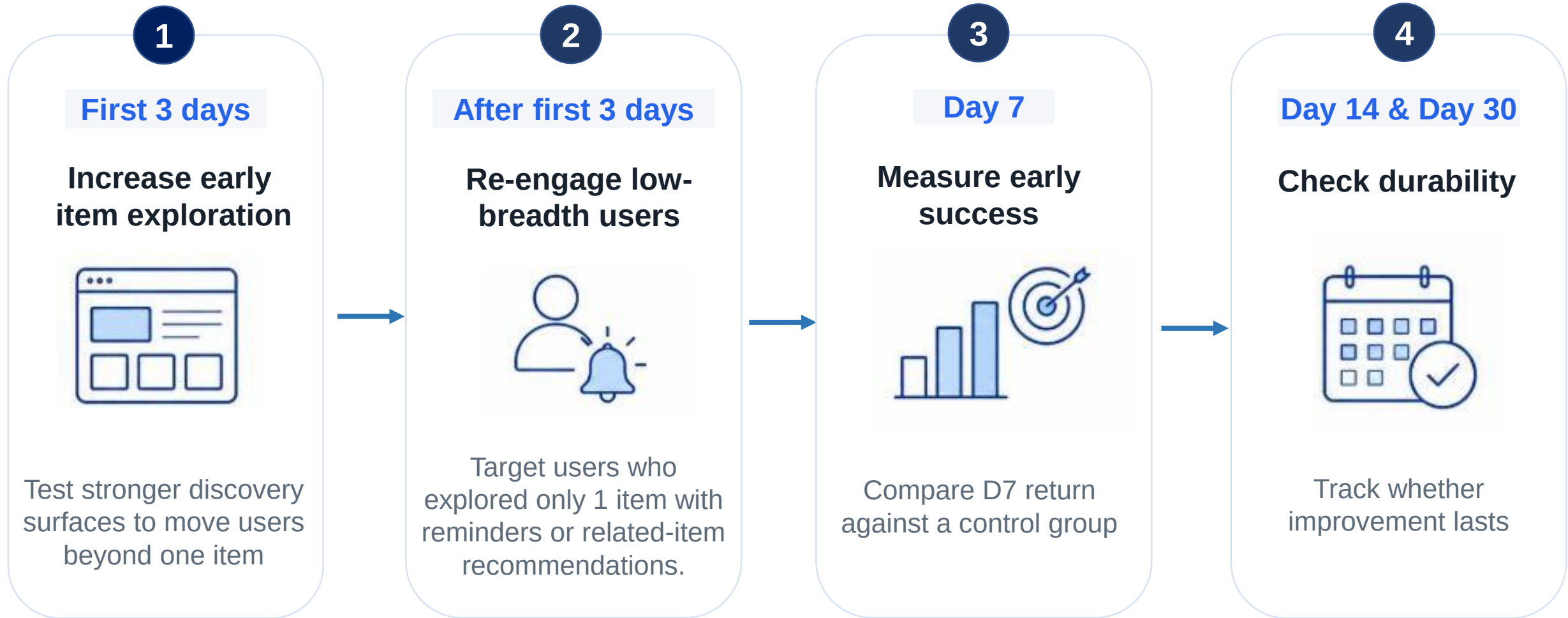


Retention is consistently ordered by early browsing breadth: users who interacted with **≥ 4 items** have the highest retention at every checkpoint, followed by **2 or 3 items**, while **1 item** users have the lowest.

This makes **early browsing breadth a useful signal** for identifying users who may need engagement support.

Note: D1 is excluded because it falls partly within the first 3 days used to define the segments.

Recommended action path



Recommendations are hypotheses to test through experiments or causal analysis, not proven causal effects.